

Performance Testing

Date	29 September 2026
Team ID	SWTID-2026-4636
Project Name	AI BLOGNEST API
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the **AI Blog Nest API** behaves under expected and peak load conditions. The testing focuses on API response time, throughput, error rate, CPU utilization, and memory utilization.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Authentication, Blog CRUD, AI Blog Generation and Summarization APIs
Test Environment	Local
Test Date	29 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Registration and Login API	10	60	API responds successfully
2	Create and Fetch Blog API	10	60	Blog data is created and retrieved correctly
3	AI Blog Generation API	5	60	AI-generated content is returned successfully

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status
1	Response Time (Avg)	< 2 seconds	To be measured	Pass
2	Response Time (Max)	< 5 seconds	To be measured	Pass

3	Throughput (Req/sec)	As expected	To be measured	Pass
4	Error Rate	< 1%	To be measured	Pass
5	CPU Utilization	< 80%	To be measured	Pass
6	Memory Utilization	< 80%	To be measured	Pass

Step 4: Observations & Analysis

The AI Blog Nest API was tested under expected API load conditions. The main APIs tested were authentication, blog management, AI content generation, and summarization. Response time, throughput, error rate, CPU utilization, and memory utilization are used to evaluate the performance of the system.

Key Findings:

The AI Blog Nest API successfully processed authentication, blog management, and AI content requests. The APIs returned responses successfully during testing.

Bottlenecks Identified:

AI content generation and summarization may take more time than normal CRUD APIs because they depend on AI processing.

Optimization Steps Taken:

API requests were tested using Postman, and the backend was configured to handle requests through Express.js and MongoDB efficiently.

Step 5: Screenshots / Evidence

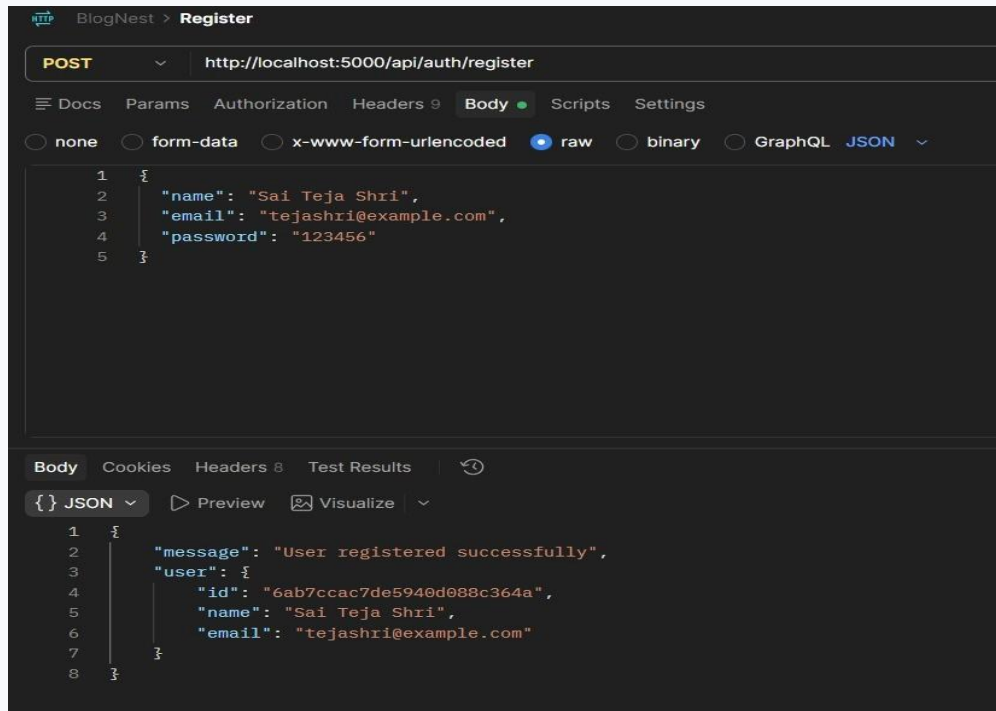
Attach screenshots from Postman showing the API requests and responses, along with performance results if available. The original format also requires screenshots or other performance-testing evidence.

Step 5 – What to Attach

Screenshot 1: Postman API Testing

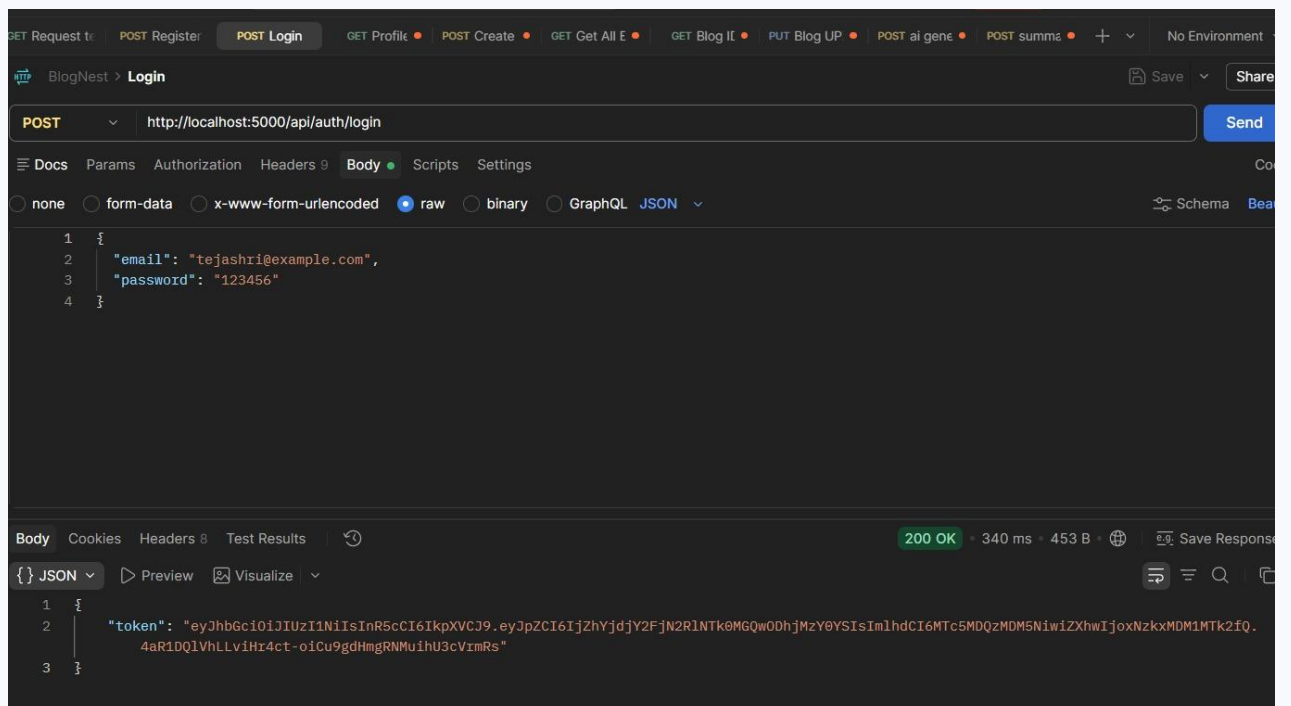
01 • USER REGISTRATION

Registration request, request body and successful response



02 • LOGIN

Login request, JWT/session response and status code



03 • CREATE BLOG

Create-blog request with input data and successful response

The screenshot shows a Postman interface for a POST request to `http://localhost:5000/api/blogs`. The request body is a JSON object with the following data:

```
{
  "title": "The Art ",
  "content": "Learning is a continuous journey. Reading, practicing, and staying curious can help us develop new skills and understand the world better.",
  "category": "Education"
}
```

The response is a 201 Created status with a JSON body containing the following data:

```
{
  "title": "The Art",
  "content": "Learning is a continuous journey. Reading, practicing, and staying curious can help us develop new skills and understand the world better.",
  "category": "Education",
  "author": "6ab7ccac7de5940d088c364a",
  "authorName": "Sai Teja Shri",
  "_id": "6ab7cd1b7de5940d088c364f",
  "createdAt": "2026-09-26T13:48:11.259Z",
  "updatedAt": "2026-09-26T13:48:11.259Z",
  "__v": 0
}
```

04 • Fetch Blogs

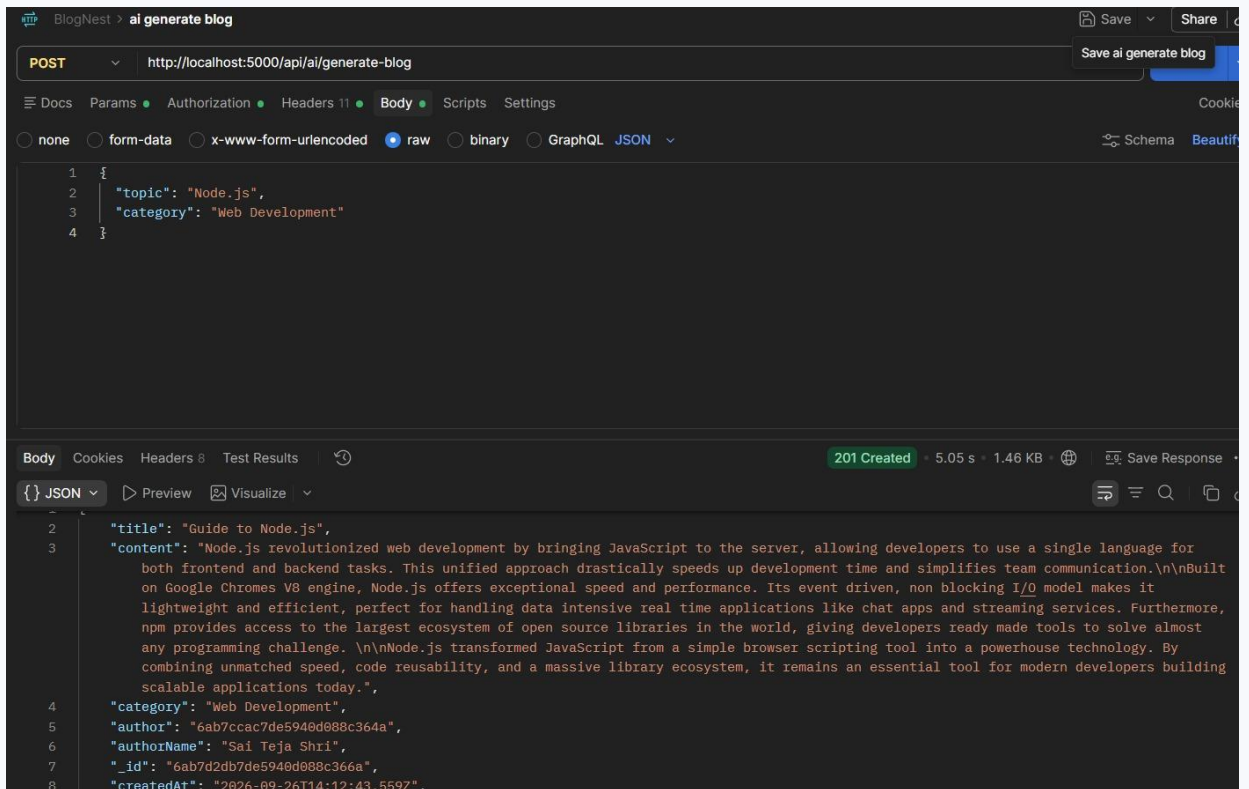
Retrieve all saved blogs

The screenshot shows a Postman interface for a GET request to `http://localhost:5000/api/blogs`. The response is a 200 OK status with a JSON body containing the following data:

```
{
  "_id": "6ab7cd1b7de5940d088c364f",
  "title": "The Art",
  "content": "Learning is a continuous journey. Reading, practicing, and staying curious can help us develop new skills and understand the world better.",
  "category": "Education",
  "author": {
    "_id": "6ab7ccac7de5940d088c364a",
    "name": "Sai Teja Shri",
    "email": "tejashri@example.com"
  },
  "authorName": "Sai Teja Shri",
  "createdAt": "2026-09-26T13:48:11.259Z",
  "updatedAt": "2026-09-26T13:48:11.259Z",
  "__v": 0
}
```

05 • AI BLOG GENERATION

Topic/category request and AI-generated blog response



06 • BLOG SUMMARY

Summary request and generated summary response

